



Defining Success Metrics for AI Infrastructure Projects

A comprehensive framework for guiding design, optimization, evaluation while balancing accuracy, efficiency, cost, and compliance in AI infrastructure projects.

Why Metrics Matter

Without clear metrics, infrastructure design becomes **pure guesswork**. Proper metrics serve as your compass for critical decisions about:

- Hardware selection
- Cloud resource allocation
- MLOps pipeline configuration

Clear KPIs demonstrate tangible value to stakeholders and are essential for your Capstone evaluation.



Core Categories of Success Metrics



Model Performance

Accuracy, F1, BLEU, ROC-AUC



System Performance

Latency, throughput, scalability



Cost Efficiency

\$/training run, \$/inference



Reliability & Security

Uptime, drift detection, IAM compliance



Sustainability

Energy use, CO₂ emissions

Model Performance Metrics

NLP Metrics

Perplexity, BLEU, ROUGE, F1 score

Computer Vision

Accuracy, IoU (Intersection over Union), precision/recall

Generative AI

Human evaluation scores, output diversity, faithfulness metrics



Each metric must directly **connect model performance to infrastructure choices**.

System Performance Metrics



Latency

Average, p95, and p99 inference time measurements

Example: **p99 < 500ms**

Throughput

Queries per second (QPS) your system can handle

Example: **500 QPS sustained**

Scalability

Elastic auto-scaling capability under varying load

Example: **10x burst capacity within 2 minutes**

Infrastructure must be tuned to meet specific SLA targets for your application.

Cost Metrics

Track expenses at every stage of the ML lifecycle:

- Training cost per epoch / per complete run
- Inference cost per request (\$/1k tokens, \$/image)
- GPU utilization efficiency percentage

Cloud billing dashboards allow real-time tracking and help identify cost optimization opportunities.





Reliability & Security Metrics



Uptime

Service availability percentage (target: 99.9%+)



Error Rates

API failures, dropped jobs, inference errors



Drift Detection

Frequency of checks + time to alert



Security Compliance

IAM audits, encryption coverage, vulnerability scanning

Sustainability Metrics

12.5 kWh

Power Draw

Per GPU node during peak training

85 kg

Carbon Footprint

CO₂ per training run

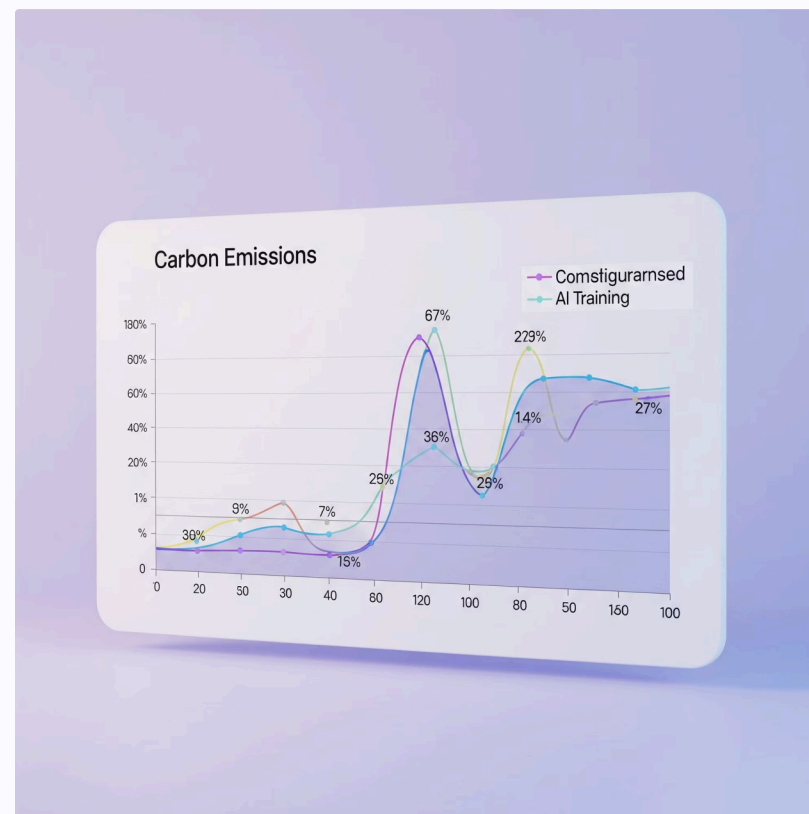
0.4 g

Inference Carbon

CO₂ per query

Measured via specialized tools:

- CodeCarbon
- MLCO₂ Calculator
- Cloud provider sustainability dashboards



Linking Metrics to Infrastructure Decisions



If latency is high

Add GPUs, implement quantization, enhance caching layer



If cost is excessive

Use spot/preemptible instances, optimize batch size



If drift is rising

Implement tighter MLOps pipeline with automated retraining

Metrics directly drive infrastructure design trade-offs



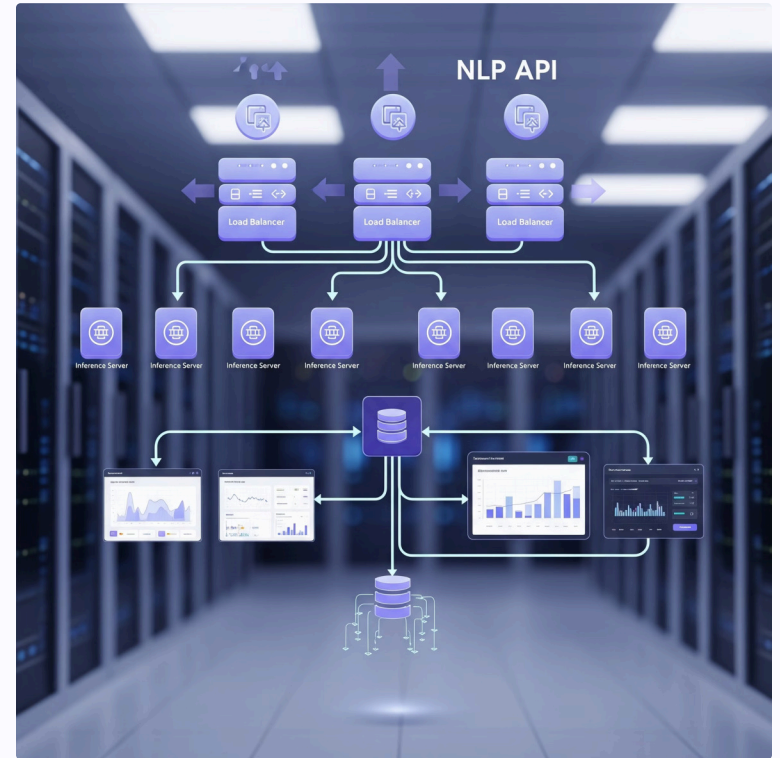
Case Study: NLP API Service

Project Goals:

- Latency < 300ms
- Accuracy > 85% F1 score
- Cost per request < \$0.001

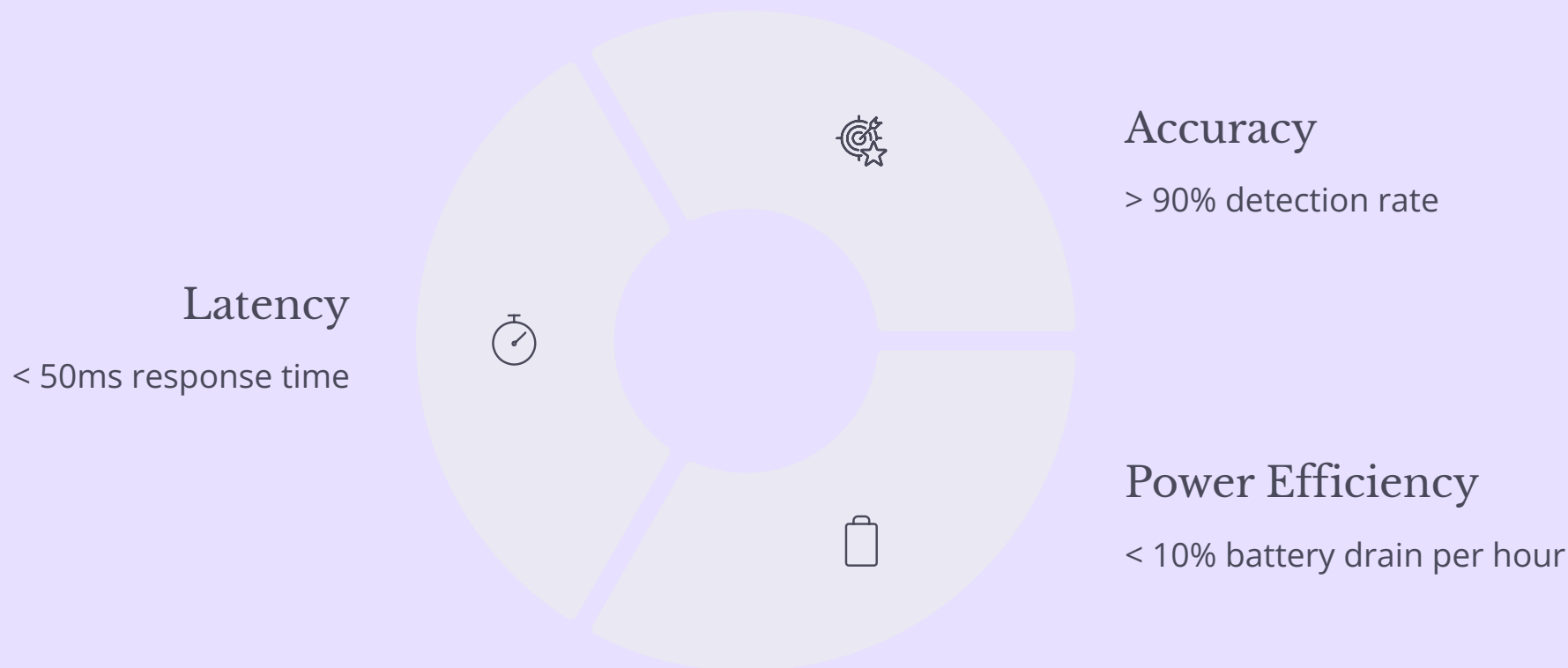
All metrics tracked via **Prometheus + MLflow** dashboards

This comprehensive tracking allowed the infrastructure team to find the optimal balance between **speed and cost**.



Case Study: Vision on Edge Devices

Goal: Real-time defect detection in factory production line



These metrics guided NPU hardware selection and quantization strategy optimization.

Case Study: Generative AI Service

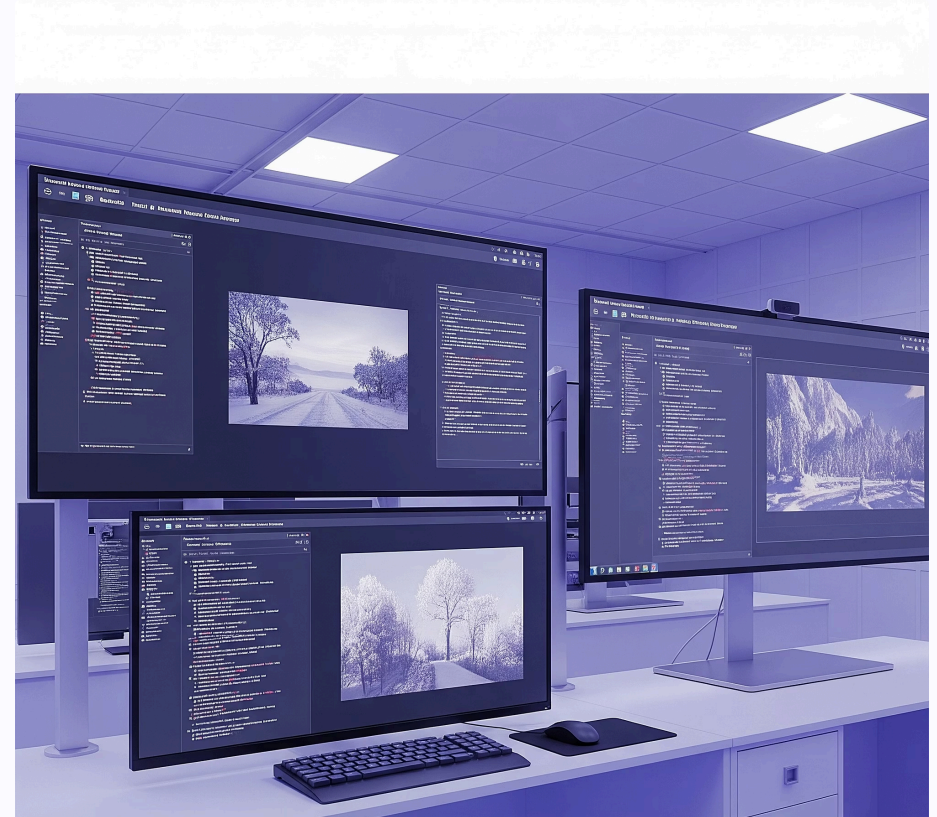
Goal: Production text-to-image model

Quality: Human-rated score > 4/5

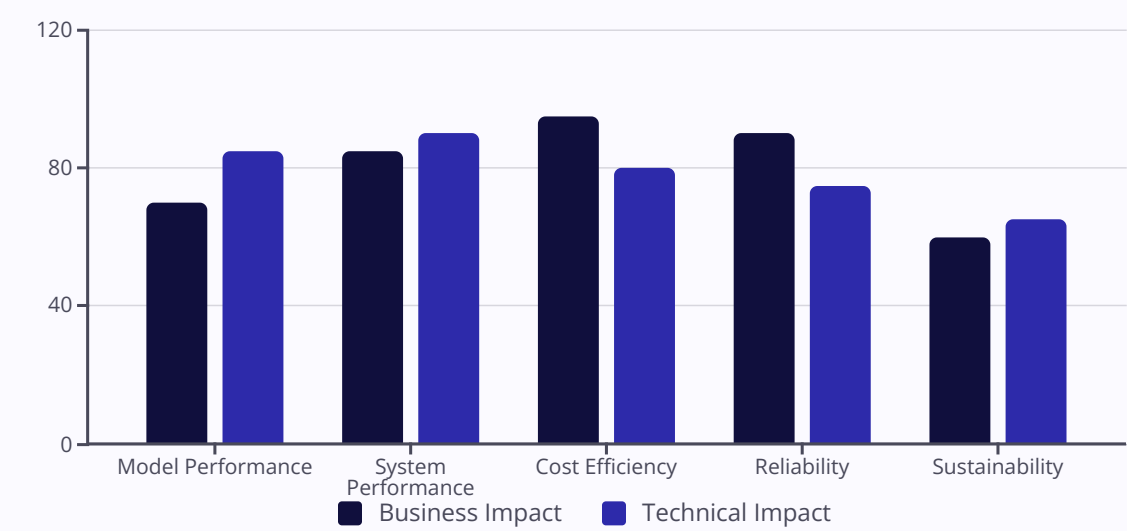
Speed: Latency < 1s per prompt

Budget: Infrastructure cost < \$10k/month

Team balanced large model capabilities with strategic inference caching to meet all requirements.



Metrics Connect AI Infrastructure to Business Value



Category	Example Metric	Tracking Tool	Business Value
Performance	Accuracy, F1, BLEU	MLflow, W&B	Product quality
System	Latency, Throughput	Prometheus, Grafana	User experience
Cost	\$/inference, GPU usage	Cloud dashboards	Profitability
Reliability	Uptime, drift detection	Kubernetes + logs	Customer trust
Sustainability	CO ₂ per run/inference	CodeCarbon	ESG goals

Key Lessons



Comprehensive Coverage

Metrics must span model, system, cost, reliability, and sustainability



Decision Drivers

Metrics should directly inform infrastructure design decisions



Measurability

Must be precisely measurable and trackable with appropriate tools



Capstone Success

Your grade depends on metrics clarity and evidence of achievement

Discussion: Prioritizing Metrics

"If you had to pick only two metrics for your Capstone..."

Latency & Cost

- Focus on operational efficiency
- Easier to measure objectively
- Directly tied to infrastructure choices
- Clear business impact (user experience, profitability)

Accuracy & Reliability

- Focus on product quality
- More directly tied to user value
- Critical for high-stakes applications
- Can differentiate from competitors

Which would you choose and why?

